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Last login: Tue Jun 16 10:54:29 on ttys001
stephenrajva@Stephens-MacBook-Air-4 ~ % cd ~/Downloads/SST_Stimulator
mkdir -p adjacent_experiment
cd adjacent_experiment
pwd
/Users/stephenrajva/Downloads/SST_Stimulator/adjacent_experiment
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % nano adjacent_sst.py
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % ls
adjacent_sst.py
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % adjacent_sst.py
zsh: command not found: adjacent_sst.py
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % sst adjacent_sst.py
Simulation is complete, simulated time: 1.5235 us
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % ls
adjacent_sst.py      adjacent_stats.csv
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % head -20 adjacent_stats.csv
ComponentName,StatisticName,StatisticSubId,StatisticType,SimTime,Rank,Sum.u64,SumSQ.u64,Count.u64,Min.u64,Max.u64
cpu,pendCycle,,Accumulator,1523500,0,16865,156659,3047,0,16
cpu,reads,,Accumulator,1523500,0,590,590,590,1,1
cpu,writes,,Accumulator,1523500,0,410,410,410,1,1
l1cache,stateEvent_GetS_I,,Accumulator,1523500,0,10,10,10,1,1
l1cache,stateEvent_GetS_S,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_GetS_M,,Accumulator,1523500,0,569,569,569,1,1
l1cache,stateEvent_GetX_I,,Accumulator,1523500,0,6,6,6,1,1
l1cache,stateEvent_GetX_S,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_GetX_M,,Accumulator,1523500,0,394,394,394,1,1
l1cache,stateEvent_GetSX_I,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_GetSX_S,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_GetSX_M,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_GetSResp_IS,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_GetXResp_IS,,Accumulator,1523500,0,10,10,10,1,1
l1cache,stateEvent_GetXResp_IM,,Accumulator,1523500,0,6,6,6,1,1
l1cache,stateEvent_GetXResp_SM,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_Inv_I,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_Inv_S,,Accumulator,1523500,0,0,0,0,0,0
l1cache,stateEvent_Inv_IS,,Accumulator,1523500,0,0,0,0,0,0
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % nano plot_stats.py
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % python3 plot_stats.py

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=== AVAILABLE STATISTICS ===
<StringArray>
[
    'pendCycle',
    'reads',
    'writes',
    'stateEvent_GetS_I',
    'stateEvent_GetS_S',
    'stateEvent_GetS_M',
    'stateEvent_GetX_I',
    'stateEvent_GetX_S',
    'stateEvent_GetX_M',
    'stateEvent_GetSX_I',
    ...
    'requests_received_PutM',
    'outstanding_requests',
    'latency_GetS',
    'latency_GetSX',
    'latency_GetX',
    'latency_Write',
    'latency_PutM',
    'cycles_with_issue',
    'cycles_attempted_issue_but_rejected',
    'total_cycles']
Length: 176, dtype: str

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=== EXTRACTED METRICS ===
Reads: 590
Writes: 410
Cache_GetS: 569
Cache_GetX: 394
PendingCycles: 16865

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Graph saved as adjacent_graph.png

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stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment %
stephenrajva@Stephens-MacBook-Air-4 adjacent_experiment % cd ~/Downloads/HPC-Upload\ 2/mpi-patterns
ls
mpi_adjacent.c mpi_halfshift.c mpi_mirrored.c
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpicc mpi_adjacent.c -o mpi_adjacent
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpirun -np 8 ./mpi_adjacent 1024
Rank 6: Sending 1024 elements to Rank 7
Rank 4: Sending 1024 elements to Rank 5
Rank 2: Sending 1024 elements to Rank 3
Rank 0: Sending 1024 elements to Rank 1
Rank 7: Received 1024 elements from Rank 6 (First element: 60)
Rank 1: Received 1024 elements from Rank 0 (First element: 0)
Rank 3: Received 1024 elements from Rank 2 (First element: 20)
Rank 5: Received 1024 elements from Rank 4 (First element: 40)
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % nano mpi_adjacent.c
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpicc mpi_adjacent.c -o mpi_adjacent
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpirun -np 8 ./mpi_adjacent 1024
Rank 2: Sending 1024 elements to Rank 3
Rank 6: Sending 1024 elements to Rank 7
Rank 0: Sending 1024 elements to Rank 1
Rank 4: Sending 1024 elements to Rank 5
Rank 7: Received 1024 elements from Rank 6 (First element: 60)
Rank 3: Received 1024 elements from Rank 2 (First element: 20)
Rank 5: Received 1024 elements from Rank 4 (First element: 40)
Rank 1: Received 1024 elements from Rank 0 (First element: 0)
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % cat adjacent_trace.csv
2,3,4096
6,7,4096
0,1,4096
4,5,4096
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % nano visualize_trace.py
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % python3 visualize_trace.py

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=== TRACE DATA ===
Source Destination Bytes
0 2 3 4096
1 6 7 4096
2 0 1 4096

```

3 4 5 4096

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Graph saved as adjacent_communication.png
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % nano mpi_adjacent.c
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % rm -f adjacent_trace.csv
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpicc mpi_adjacent.c -o mpi_adjacent
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpirun -np 8 ./mpi_adjacent 1024
Rank 6: Sending 1024 elements to Rank 7
Rank 0: Sending 1024 elements to Rank 1
Rank 2: Sending 1024 elements to Rank 3
Rank 4: Sending 1024 elements to Rank 5
Rank 5: Received 1024 elements from Rank 4 (First element: 40)
Rank 7: Received 1024 elements from Rank 6 (First element: 60)
Rank 3: Received 1024 elements from Rank 2 (First element: 20)
Rank 1: Received 1024 elements from Rank 0 (First element: 0)
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % cat adjacent_trace.csv
4,5,4096,0.000067
6,7,4096,0.000081
2,3,4096,0.000127
0,1,4096,0.000167
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % nano visualize_trace.py
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % python3 visualize_trace.py
```

```
=== TRACE DATA ===
Source Destination Bytes Latency
0 4 5 4096 0.000067
1 6 7 4096 0.000081
2 3 4096 0.000127
3 0 1 4096 0.000167
```

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Graph saved as adjacent_latency.png
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpirun -np 8 ./mpi_adjacent 100
Rank 6: Sending 100 elements to Rank 7
Rank 0: Sending 100 elements to Rank 1
Rank 4: Sending 100 elements to Rank 5
Rank 2: Sending 100 elements to Rank 3
Rank 7: Received 100 elements from Rank 6 (First element: 60)
Rank 3: Received 100 elements from Rank 2 (First element: 20)
Rank 5: Received 100 elements from Rank 4 (First element: 40)
Rank 1: Received 100 elements from Rank 0 (First element: 0)
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpirun -np 8 ./mpi_adjacent 10000
Rank 6: Sending 10000 elements to Rank 7
Rank 0: Sending 10000 elements to Rank 1
Rank 2: Sending 10000 elements to Rank 3
Rank 4: Sending 10000 elements to Rank 5
Rank 3: Received 10000 elements from Rank 2 (First element: 20)
Rank 1: Received 10000 elements from Rank 0 (First element: 0)
Rank 5: Received 10000 elements from Rank 4 (First element: 40)
Rank 7: Received 10000 elements from Rank 6 (First element: 60)
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpirun -np 8 ./mpi_adjacent 100000
Rank 2: Sending 100000 elements to Rank 3
Rank 3: Received 100000 elements from Rank 2 (First element: 20)
Rank 6: Sending 100000 elements to Rank 7
Rank 4: Sending 100000 elements to Rank 5
Rank 0: Sending 100000 elements to Rank 1
Rank 7: Received 100000 elements from Rank 6 (First element: 60)
Rank 5: Received 100000 elements from Rank 4 (First element: 40)
Rank 1: Received 100000 elements from Rank 0 (First element: 0)
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % rm -f adjacent_trace.csv
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % python3 visualize_trace.py
Traceback (most recent call last):
  File "/Users/stephenrajva/Downloads/HPC-Upload 2/mpi-patterns/visualize_trace.py", line 5, in <module>
    df = pd.read_csv(
          "adjacent_trace.csv",
          header=None,
          names=["Source", "Destination", "Bytes", "Latency"]
    )
  File "/opt/homebrew/lib/python3.14/site-packages/pandas/io/parsers/readers.py", line 873, in read_csv
    return _read(filepath_or_buffer, kwds)
  File "/opt/homebrew/lib/python3.14/site-packages/pandas/io/parsers/readers.py", line 300, in _read
    parser = TextFileReader(filepath_or_buffer, **kwds)
  File "/opt/homebrew/lib/python3.14/site-packages/pandas/io/parsers/readers.py", line 1645, in __init__
    self._engine = self._make_engine(f, self._engine)
                    ~~~~~^~~~~~
  File "/opt/homebrew/lib/python3.14/site-packages/pandas/io/parsers/readers.py", line 1904, in _make_engine
    self._handles = get_handle(
                    ~~~~~^~~~~
    f,
    ^^
    ...<6 lines>...
    storage_options=self.options.get("storage_options", None),
    ~~~~~^~~~~~
    )
    ^
  File "/opt/homebrew/lib/python3.14/site-packages/pandas/io/common.py", line 930, in get_handle
    handle = open(
        handle,
        ...<3 lines>...
        newline="",
    )
FileNotFoundError: [Errno 2] No such file or directory: 'adjacent_trace.csv'
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % rm -f adjacent_trace.csv
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpirun -np 8 ./mpi_adjacent 100000
Rank 0: Sending 100000 elements to Rank 1
Rank 2: Sending 100000 elements to Rank 3
Rank 4: Sending 100000 elements to Rank 5
Rank 6: Sending 100000 elements to Rank 7
Rank 7: Received 100000 elements from Rank 6 (First element: 60)
Rank 1: Received 100000 elements from Rank 0 (First element: 0)
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Rank 3: Received 100000 elements from Rank 2 (First element: 20)
```

```
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % python3 visualize_trace.py
```

```
=== TRACE DATA ===
```

	Source	Destination	Bytes	Latency
0	0	1	400000	0.000741
1	6	7	400000	0.000490
2	2	3	400000	0.000979
3	4	5	400000	0.000959

Graph saved as adjacent_latency.png

```
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % mpirun -np 8 ./mpi_adjacent 100000
```

Rank 2: Sending 100000 elements to Rank 3

Rank 3: Received 100000 elements from Rank 2 (First element: 20)

Rank 6: Sending 100000 elements to Rank 7

Rank 4: Sending 100000 elements to Rank 5

Rank 0: Sending 100000 elements to Rank 1

Rank 5: Received 100000 elements from Rank 4 (First element: 40)

Rank 7: Received 100000 elements from Rank 6 (First element: 60)

Rank 1: Received 100000 elements from Rank 0 (First element: 0)

```
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % cat adjacent_trace.csv
```

0,1,400000,0.000741

6,7,400000,0.000490

2,3,400000,0.000979

4,5,400000,0.000959

2,3,400000,0.000182

4,5,400000,0.000086

6,7,400000,0.000178

0,1,400000,0.000186

```
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns % python3 visualize_trace.py
```

```
=== TRACE DATA ===
```

	Source	Destination	Bytes	Latency
0	0	1	400000	0.000741
1	6	7	400000	0.000490
2	2	3	400000	0.000979
3	4	5	400000	0.000959
4	2	3	400000	0.000182
5	4	5	400000	0.000086
6	6	7	400000	0.000178
7	0	1	400000	0.000186

Graph saved as adjacent_latency.png

```
stephenrajva@Stephens-MacBook-Air-4 mpi-patterns %
```